

Analyser — Developer Guide

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This guide covers the technical internals of the Analyser application for contributors and maintainers. It focuses on areas not already covered by inline code comments.

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1. Project Setup

```

npm install
npm run dev          # dev server on http://localhost:5173
npm run build        # production build – also generates stats.html bundle report
npm run check        # svelte-check + tsc
npm test             # vitest unit tests

```

`npm run build` generates `stats.html` (via `rollup-plugin-visualizer`) in the project root. Open it in a browser after any build to inspect chunk sizes and identify modules that have regressed into the initial bundle. The file is gitignored.

The app uses **SvelteKit** with the **Vercel adapter** (`@sveltejs/adapter-vercel`). Deployment is to Vercel via GitHub Actions on push to `main`.

2. Architecture Overview

```

src/
├─ lib/
│  ├─ fit/          # FIT file parsing (Web Worker, normalisation, device classification)
│  ├─ align/        # Activity alignment (timestamp sync, distance interpolation)
│  │               # - distance.ts: interpolateToDistanceAxis, distanceStep(totalDistance, distanceStep)
│  │               #   distanceStep: ≤50 km → 10 m step; >50 km → 20 m step (halved)
│  ├─ analytics/    # Smoothing, mean/max curves, summary statistics, anomaly detection
│  │               # - anomalies.ts: detectAnomalies(), groupAnomalyEvents(), groupAnomalyEventsByChannel()
│  │               # - stats.ts: mean(), stdDev() – population statistics ignoring outliers
│  │               # - downsample.ts: downsampleGps(points, maxPoints, epsilon) –
│  │               #   simplification; GPS_MAX_POINTS=500 hard cap; iterative RDP (Recursive
│  │               #   no recursion limit) with uniform-decimate fallback if RDP fails
│  ├─ compare/      # Delta computation, segment analysis
│  ├─ export/       # Client-side data export
│  │   ├─ columns.ts # Shared column utilities: CHANNEL_KEYS, presentChannelKeys
│  │   │             #   formatCellValue(), escapeCsvField(); ExportFormat
│  │   ├─ csv.ts     # buildCsv(activities) → RFC 4180 string; single-file export
│  │   │             #   multi-file prepends it; CRLF line endings; channel order
│  │   ├─ excel.ts   # buildWorkbook(activities) → ArrayBuffer (.xlsx via ExcelJS)
│  │   ├─ exportActivities.ts # exportActivities(activities, format): lazy-loads csv
│  │   │             #   triggers download; format defaults to 'csv'
│  │   └─ download.ts # triggerDownload(), downloadPng(), localDateString()
│  └─ components/
│     ├─ charts/    # ECharts wrappers (ECharts is lazy-loaded – see §2.1)
│     │             # - echarts-loader.ts: singleton loadECharts() – one dynamic import
│     │             # - chart-skeleton.css: shared shimmer skeleton styles used by
│     │             # - TimeSeriesChart.svelte: per-series stats row (min/avg/max,
│     │             #   All line series use sampling: 'lttb' (Largest-Triangle-Three-Corner)
│     │             #   automatic ECharts-side downsampling on large datasets (>2000 points)
│     │             # - TimeSeriesChartUtils.ts: computeSeriesStats(), formatStats()
│     │             # - StripChart.svelte: metric strip chart wrapper (map tab)

```

```

| | | # - StripToggle.svelte: line/gradient pill toggle
| | | # - StripChart.utils.ts: shouldShowGradient(), GRADIENT_COLOUR_
| | | # - png-btn.css: shared PNG download button styles
| | └─ map/ # Leaflet map + ActivityMap.component.test.ts (jsdom)
| | | # - ActivityMap.utils.ts: extractGpsPoints, metricValuesForGpsP
| | | #   smoothedValues) – maps downsampled GPS points to metric val
| | | # - DeviceToggleBar.utils.ts: buildAnomalyCounts(activities) →
| | | #   aggregates event-level anomaly counts across all activities
| | └─ ui/ # Layout components, controls, SyncPanel
| └─ server/
|   └─ redis.ts # Upstash Redis singleton (server-only)
| └─ stores/
|   └─ deviceLabels.ts # Device label persistence (localStorage)
|   └─ sync.ts # Cross-device sync logic
|   └─ session.ts # Activity session state; activityColourMap assigns stat
|   └─ toast.ts # Transient toast notification store (addToast, removeTo
|   └─ viewport.ts # Responsive breakpoint store
| └─ utils/ # Pure utility functions
|   # - binarySearch.ts: lowerBound<T>(arr, key, target) – shared 1
|   # - channels.ts: deriveAvailableChannels()
|   # - deviceChannels.ts: deviceKey(), deriveDeviceLabel(), groups
|   # - fetchWithRetry.ts: fetchWithRetry(), isRetryable(), compute
|   # - formatAge.ts: formatAge(ts) → human-readable relative time
|   # - formatting.ts: formatPace(decimalMinutes) → "M:SS" string -
|   # - indoorWarnings.svelte.ts: createIndoorWarnings() composable
|   # - lapMarkers.ts: buildLapMarkers()
|   # - segments.ts: buildSegments()
| └─ validation.ts # Shared regex constants (UUID, short code)
| └─ types.ts # Domain types
|   # - Anomaly, AnomalyType, DetectionStrategy, AnomalyDetectionOp
|   # - ChannelKey now includes 'position' (GPS-only; no numeric se
|   # - GpsPointWithDistance: lat, lon, distance, recordIndex (0(1)
|   # - GpsPointWithMetric extends GpsPointWithDistance: metricValu
└─ routes/
  └─ api/
    └─ labels/
      └─ [uuid]/+server.ts # GET / PUT label store
      └─ resolve/[code]/+server.ts # GET short-code resolver
    └─ export-btn.css # Shared CSS for the Export Data button in tab bars
    └─ map-panel.css # Shared CSS for Map tab layout (.map-wrap--has-strip, .stri
    └─ +layout.svelte # App shell, initSync, ?sync= param
    └─ +page.svelte # Landing / drop zone
    └─ compare/+page.svelte
    └─ event/+page.svelte

```

2.1 ECharts Lazy-Loading

ECharts (≈ 1.1 MB / 374 kB gzipped) is **not in the initial bundle**. It is deferred to a separate async chunk and fetched only when the first chart component mounts.

How it works

All four chart components (`TimeSeriesChart` , `DeltaChart` , `MeanMaxChart` , `SegmentChart`) import ECharts through a shared singleton loader:

```
// src/lib/components/charts/echarts-loader.ts
import type * as EChartsNamespace from 'echarts';
export type EChartsModule = typeof EChartsNamespace;

let cached: Promise<EChartsModule> | null = null;

export function loadECharts(): Promise<EChartsModule> {
  if (!cached) {
    cached = import('echarts') as Promise<EChartsModule>;
  }
  return cached;
}
```

Each chart component calls `loadECharts()` inside an `async onMount` :

```
onMount(async () => {
  ec = await loadECharts(); // dynamic import (first call fetches; subsequent calls return cached)
  chart = ec.init(container, undefined, { renderer: 'canvas' });
  chart.setOption(buildOption(), { notMerge: true }); // explicit initial render
  // ... event binding, ResizeObserver, etc.
  ready = true;
});
```

Why `chart.setOption()` is called explicitly in `onMount`

Three of the four chart components (`TimeSeriesChart` , `DeltaChart` , `SegmentChart`) declare `chart` as a plain `let` variable rather than `$state` . Because `chart` is not reactive, the Svelte `$effect` that normally calls `chart?.setOption()` fires *before* the `await loadECharts()` resolves — hitting a no-op. The explicit `setOption` call immediately after `ec.init()` guarantees the first render regardless of reactive timing.

`MeanMaxChart` uses `let chart = $state<ECharts | undefined>(undefined)` — assigning the chart instance triggers the `$effect` reactively — but still includes the explicit `setOption` call for consistency and to make all four components follow the same pattern.

Loading skeleton

While `loadECharts()` is in flight the chart canvas (`bind:this={container}`) stays in the DOM with `visibility: hidden` so `ec.init()` can be called on an element with real dimensions. A shimmer placeholder div replaces the visual space:

```
{#if !ready}
  <div class="chart-canvas chart-skeleton" aria-hidden="true"></div>
```

```
{/if}
```

```
<div bind:this={container} class="chart-canvas" style:visibility={ready ? 'visible' :
```

Skeleton styles live in `src/lib/components/charts/chart-skeleton.css` (imported by all four chart components) and use CSS custom properties defined in `src/routes/layout.css` :

```
/* dark theme (default) */
--skeleton-from: #1e293b;
--skeleton-to:   #334155;

/* light theme */
--skeleton-from: #e2e8f0;
--skeleton-to:   #f1f5f9;
```

Bundle verification

`rollup-plugin-visualizer` is configured in `vite.config.ts`. Every `npm run build` writes `stats.html` to the project root (gitignored). Open it to verify ECharts appears only as a dynamic entry (`isDynamicEntry: true` in the Vite manifest) and is absent from all entry-point sync dependency trees.

Adding a new chart component

If you add a fifth chart component:

1. Import from the shared loader: `import { loadECharts, type EChartsModule } from './echarts-loader'`
2. Keep `import type { ECharts, EChartsOption } from 'echarts'` — type-only imports are erased at compile time and do not create a bundle dependency
3. Use `async onMount`, call `loadECharts()`, call `chart.setOption(buildOption(), { notMerge: true })` immediately after `ec.init()`
4. Import `chart-skeleton.css` and add the skeleton/visibility template pattern

3. Cross-Device Label Sync

PR #80 added cross-device sync for device labels. Users rename sensor pills (e.g. "Device 1" → "Polar H10") and those names are shared across all their browsers and devices without requiring an account.

3.1 Design Goals

- **Account-free** — identity is a randomly generated UUID stored in `localStorage`, not tied to any login.
- **Cloud-authoritative** — the Upstash Redis store is treated as the source of truth; local state is overwritten on pull.
- **Fire-and-forget pushes** — label changes auto-push in the background; errors surface in the UI but never block local usage.

- **Human-readable codes** — a short 8-character code (e.g. `E6Y-NXEMF`) derived from the UUID lets users type the code on another device instead of scanning a QR.
- **90-day TTL** — entries in Redis expire after 90 days of inactivity with no explicit deletion needed.

3.2 Data Model in Redis

Two key types are stored:

Key pattern	Value	TTL
<code>labels:{uuid}</code>	JSON object: <code>{ [antDeviceKey]: labelString }</code>	90 days, rolling
<code>code:{shortCode}</code>	UUID string	90 days, rolling

Both keys are refreshed together on every PUT, so activity on one extends the other's TTL automatically.

The `antDeviceKey` format is defined by `deviceKey()` in `src/lib/utils/deviceChannels.ts` — it encodes manufacturer, product ID, and device type into a stable string used as the label map key.

3.3 Identity Lifecycle

First visit:

```
crypto.randomUUID() → uuid
deriveShortCode(uuid) → shortCode
localStorage: SYNC_ID_KEY, SYNC_CODE_KEY
PUT /api/labels/{uuid} → seed remote with current labels
```

Returning visit:

```
Read uuid / shortCode from localStorage
GET /api/labels/{uuid} → pull remote labels → replaceAllLabels()
```

Label change (any visit):

```
setOnLabelChange hook → pushLabels(uuid, shortCode) [fire and forget]
```

Adopt external identity (QR / copy-link / short code):

```
adoptSyncIdentity(uuid) → store uuid in localStorage → pullLabels(uuid)
```

Reset identity:

```
crypto.randomUUID() → new uuid
localStorage cleared of old identity
pushLabels(newUuid, newShortCode) → seed remote under new identity
```

3.4 Sync Store (`sync.ts`)

Location: `src/lib/stores/sync.ts`

Exports:

Export	Type	Description
<code>syncStatus</code>	<code>Writable<SyncStatus></code>	Reactive store for UI state (uuid, shortCode, lastSynced, error, syncing)
<code>initSync()</code>	<code>async () => (() => void) undefined</code>	Idempotent — safe to call multiple times; second call is a no-op. Returns a cleanup function on first call (call in <code>onDestroy</code> to deregister hook and allow HMR reinit); returns <code>undefined</code> if already initialised. SSR-safe.
<code>pushLabels(uuid, shortCode)</code>	<code>async</code>	Upload current labels to Redis. Retries up to 3× on transient errors. Sets <code>syncStatus.syncing</code> while in progress. Errors update <code>syncStatus.error</code> .
<code>pullLabels(uuid)</code>	<code>async</code>	Download labels from Redis. Retries up to 3× on transient errors. Sets <code>syncStatus.syncing</code> while in progress.
<code>resolveCode(code)</code>	<code>async → uuid</code>	Resolve short code to UUID via API. Retries up to 3× on transient errors. Throws on 404.
<code>adoptSyncIdentity(uuid)</code>	<code>async</code>	Replace local identity with external UUID and pull.
<code>resetSyncIdentity()</code>	<code>async</code>	Generate fresh UUID, push current labels under it.
<code>deriveShortCode(uuid)</code>	<code>string</code>	Pure function: BigInt base-36 encode of UUID, 8 chars, <code>XXX-XXXXX</code> .
<code>getSyncStatus()</code>	<code>SyncStatus</code>	Non-reactive snapshot (for use outside Svelte components).

`SyncStatus` shape:

```
type SyncStatus = {
  uuid:      string | null;
  shortCode: string | null;
  lastSynced: string | null; // ISO timestamp
  error:     string | null;
  syncing:   boolean;        // true while any sync operation is in flight (including
};
```

Location: `src/lib/utils/fetchWithRetry.ts`

All three sync network calls (`pushLabels` , `pullLabels` , `resolveCode`) use a shared retry utility rather than bare `fetch` :

```
fetchWithRetry(input, init?, opts?): Promise<Response>
```

Default options: `maxRetries: 3` , `baseDelayMs: 1000` , `jitter: 0.25` (±25%).

Retry classification (`isRetryable`):

Response / Error	Retried?
Network error (<code>TypeError</code>)	☐ Yes
5xx	☐ Yes
429	☐ Yes
4xx (except 429)	☒ No
2xx / 3xx	☒ No

Backoff schedule (zero jitter, for clarity):

Attempt	Delay
0 → 1	~1 s
1 → 2	~2 s
2 → 3	~4 s

When the server returns a `429` with a `Retry-After` header, that value (in seconds) overrides the exponential delay for that attempt, preventing wasted retries within the rate-limit window.

After all retries are exhausted: the last `Response` is returned for HTTP errors; the last network error is thrown.

Test isolation: `_setSleepFn(fn)` replaces the internal sleep function so that retry tests do not incur real delays (used in `sync.test.ts` via `beforeEach`).

3.4b Sidebar Sync Indicator

Location: `src/lib/components/ui/Sidebar.svelte`

A persistent status indicator is always visible in the sidebar footer, showing the current sync state without requiring the user to expand the `SyncPanel`. It subscribes to `$syncStatus` via three `$derived.by()` values:

State	Condition	Icon	Colour
syncing	<code>\$syncStatus.syncing === true</code>	Animated two-arc spinner SVG	Blue (<code>#3b82f6</code>)
error	<code>\$syncStatus.error !== null</code>	Triangle warning SVG	Amber (<code>#fbbf24</code>)
ok	<code>\$syncStatus.lastSynced !== null</code>	Filled circle SVG	Green (<code>#10b981</code>)
muted	All others (setting up)	Filled circle SVG	<code>--color-muted</code>

The element carries `aria-live="polite"` and a dynamic `aria-label` (e.g. "Synced just now", "Sync error: ...") for screen readers.

When `syncing === true` and `error !== null` simultaneously (i.e. auto-retry is in progress after a previous failure), `SyncPanel.svelte` shows "Retrying..." in blue instead of the normal amber "⚠ Sync error – retry" link.

3.5 DeviceLabels Integration

Location: `src/lib/stores/deviceLabels.ts`

Three additions support sync:

Function	Description
<code>getAllLabels()</code>	Returns a snapshot <code>Record<string, string></code> of all stored labels
<code>replaceAllLabels(map)</code>	Overwrites <code>localStorage</code> with the provided map and notifies the store
<code>setOnLabelChange(fn null)</code>	Registers a callback invoked after every <code>setDeviceLabel</code> / <code>removeDeviceLabel</code> . Pass <code>null</code> to deregister.

Only one change callback can be registered at a time. `initSync` registers its push callback after the initial push/pull, so subsequent label changes auto-push. The cleanup function returned by `initSync` calls `setOnLabelChange(null)` to deregister on teardown.

3.6 Toast Notification Store

Location: `src/lib/stores/toast.ts`

Component: `src/lib/components/ui/ToastContainer.svelte`

A lightweight, reusable toast system for surfacing transient non-blocking messages to the user. The store is framework-agnostic — any module can call `addToast` without importing Svelte components.

API

Export	Signature	Description
<code>toasts</code>	<code>Readable<Toast[]></code>	Reactive list of active toasts. Read-only — prevents external mutation.
<code>addToast</code>	<code>(message: string, level?: ToastLevel, duration?: number) => void</code>	Adds a toast. Default level: <code>'warning'</code> . Default duration: <code>5000 ms</code> .
<code>removeToast</code>	<code>(id: number) => void</code>	Removes a toast immediately and clears its auto-dismiss timer.

Types

```
type ToastLevel = 'info' | 'warning' | 'error';

interface Toast {
  id: number;
  message: string;
  level: ToastLevel;
}
```

Behaviour

- Each toast is assigned a unique auto-incrementing `id`.
- `addToast` schedules a `setTimeout` to call `removeToast(id)` after `duration` ms. The timer handle is stored in a `Map<id, timer>` and cleared by `removeToast` to prevent orphaned timers when a toast is dismissed early.
- `removeToast` with an unknown `id` is a safe no-op.

ToastContainer component

`ToastContainer.svelte` subscribes to `$toasts` and renders the most recent `MAX_VISIBLE` (3) toasts as a fixed-position stack (bottom-right on desktop; full-width above the bottom nav bar on phone). It is mounted **once** in `+layout.svelte` so toasts are visible on every page.

Each toast has a level-appropriate left-border accent and dismiss button (×). Colours use hardcoded semantic values (amber/blue/red) against `--color-card` / `--color-text` backgrounds, ensuring correct appearance in both light and dark themes.

Current callers

Caller	Level	Message
deviceLabels.ts — saveLabels catch	warning	'Device label could not be saved – storage full' (QuotaExceededError) or 'Device label could not be saved' (other errors)
DropZone.svelte — parse error catch (single-file and multi-file)	error	'<filename>: <parser error message>' — fires alongside the existing inline error text so the error persists after navigation
parser.ts — normalise() negative elapsed	warning	'"<filename>": N record(s) with negative elapsed time were removed.'
parser.ts — normalise() out-of-order records	warning	'"<filename>": records were out of order and have been sorted automatically.'

3.7 API Routes

See the full API reference in [docs/api-reference.md](#) .

GET /api/labels/[uuid] — returns `{ labels }` or 404.

PUT /api/labels/[uuid] — expects `{ labels: Record<string,string>, shortCode: string }` . Writes both `labels:{uuid}` and `code:{shortCode}` to Redis with a 90-day TTL.

GET /api/labels/resolve/[code] — resolves short code to UUID. Rate-limited at 10 req/min/IP via Redis `INCR / EXPIRE` pipeline (globally enforced across all Vercel instances). Responds 429 with `Retry-After` header on limit exceeded; fails open if Redis is unavailable.

All routes validate inputs against `UUID_REGEX` / `SHORT_CODE_REGEX` from `src/lib/validation.ts` and return structured JSON errors on 400/404/429/500.

3.8 SyncPanel Component

Location: `src/lib/components/ui/SyncPanel.svelte`

A collapsible panel rendered in the Sidebar footer. It is always mounted (not conditionally rendered) so the sync toggle row is always visible.

Key internal state:

State	Type	Purpose
<code>isOpen</code>	<code>boolean</code>	Controls panel expansion
<code>codeInput</code>	<code>string</code>	Bound to the short-code text input
<code>codeError</code> / <code>codeSuccess</code>	<code>string null</code> / <code>boolean</code>	Inline validation feedback
<code>isResolving</code>	<code>boolean</code>	Disables input during async resolve
<code>copied</code>	<code>boolean</code>	Briefly true after clipboard write, drives button label

Derived values:

- `syncUrl` — `${origin}?sync=${uuid}`, used for both QR generation and copy-link.
- `qrSvg` — generated client-side via `uqr` (`renderSVG(syncUrl)`). No server round-trip.

The component validates code input against `SHORT_CODE_REGEX` before hitting the API, providing immediate feedback on format errors.

3.9 Layout Wiring

`src/routes/+layout.svelte`

Three sync-related wiring points:

1. `onMount` calls `initSync()` — idempotent; safe if called more than once (e.g. during HMR). Stores the returned cleanup function in `cleanupSync`.
2. `onDestroy` calls `cleanupSync?.()` — deregisters the label-change hook and resets the initialised flag, enabling a clean reinit if the layout remounts (HMR).
3. URL parameter handling: if the page loads with `?sync={uuid}`, `adoptSyncIdentity(uuid)` is called and the param is stripped from the URL using `history.replaceState` to avoid sharing it inadvertently via browser history.

3.10 Validation

`src/lib/validation.ts`

Shared between API routes (server) and `SyncPanel` (client) to avoid duplication:

```
UUID_REGEX      = /^[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}$/i
SHORT_CODE_REGEX = /^[A-Z0-9]{3}-[A-Z0-9]{5}$/i
```

3.11 Environment Variables

Variable	Required	Description
<code>UPSTASH_REDIS_REST_URL</code>	Yes	REST endpoint for your Upstash Redis database
<code>UPSTASH_REDIS_REST_TOKEN</code>	Yes	Read-write token for the database

These are read via `$env/dynamic/private` (SvelteKit server-only) in `src/lib/server/redis.ts`. They must be set in:

- **Local development:** `.env.local` (git-ignored, never committed)
- **Vercel:** Project Settings → Environment Variables

The Redis singleton (`getRedis()`) throws an explicit error if credentials are absent, which the API routes catch and surface as HTTP 500. This prevents silent failures.

3.12 Local Development Setup

1. Create an [Upstash](#) account and create a Redis database (free tier is sufficient).
2. Copy the REST URL and token from the Upstash console.
3. Create `.env.local` in the project root:

```
UPSTASH_REDIS_REST_URL=https://your-db.upstash.io
```

```
UPSTASH_REDIS_REST_TOKEN=your-token-here
```

4. Run `npm run dev` — sync features are fully functional locally.

Tip: If you do not configure Redis, the app still works — sync API calls will return 500 and `syncStatus.error` will be set, but all FIT file analysis features continue to function normally.

3a. Map Tab — Metric Strip Chart

PR #82 added a co-visible metric strip chart to the Map tab on both `/compare` and `/event` pages. When the user selects a metric in the **Colour by** picker, a chart for that channel slides in below the map with full bidirectional hover sync.

Component hierarchy

```
compare/+page.svelte (or event/+page.svelte)
├── .map-panel (flex column)
│   ├── .map-wrap (flex: 7)
│   │   └── <ActivityMap
│   │       ├── onMetricChannelChange={ch => mapMetricChannel = ch}
│   │       ├── hoveredDistance={stripHoveredDistance}
│   │       └── />
│   └── .strip-wrap (flex: 3, shown only when mapMetricChannel !== null)
│       └── <CollapsiblePanel> (collapses on tablet/phone)
│           └── <StripChart
│               ├── channel={mapMetricChannel}
│               ├── seriesInputs={stripSeriesInputs}
│               ├── lapMarkers={...}
│               ├── onHoverDistance={handleStripHoverDistance}
│               └── externalHoverDistance={mapStripHoveredDistance}
│           └── />
```

Hover sync

The strip chart uses a **second, independent hover loop** (`stripHoveredDistance` / `mapStripHoveredDistance`), completely separate from the existing Charts-tab ↔ Map loop (`chartHoveredDistance` / `mapHoveredDistance`). This separation is intentional:

- The existing loop is **gated on** `$xAxisMode === 'distance'` — it only fires when the Charts tab is in distance mode.
- The strip loop is **never gated** — the strip chart always operates in distance mode (`forceDistanceAxis={true}`), so the sync is always valid regardless of the global x-axis setting.

Key design points

Concern	Approach
Strip always uses distance axis	<code>forceDistanceAxis={true}</code> prop on <code>TimeSeriesChart</code> inside <code>StripChart</code>
Isolated zoom/pan	<code>groupId="map-strip"</code> — separate ECharts connect group from "compare-charts" / "event-charts"
Gradient mode	<code>shouldShowGradient()</code> utility in <code>StripChart.utils.ts</code> ; single-file only (multi-file stays as lines)
Shared layout CSS	<code>.map-wrap--has-strip</code> and <code>.strip-wrap</code> styles extracted to <code>src/routes/map-panel.css</code> and imported in both pages
Magic-string safety	Gradient sentinel value is <code>GRADIENT_COLOUR_TOKEN = '__gradient__'</code> (exported constant, not inline string)

`ActivityMap` — `onMetricChannelChange` callback

`ActivityMap.svelte` accepts an optional `onMetricChannelChange?: (channel: ChannelKey | null) => void` prop. It fires via a `$effect` whenever the internal `metricChannel` state changes (on picker selection and on file change/reset). Existing callers that do not pass this prop are unaffected.

3b. Data Export

PR #83 added Excel workbook and per-chart PNG downloads. PR #111 added CSV export and extracted shared column utilities into `columns.ts`.

Module layout

File	Responsibility
<code>src/lib/export/columns.ts</code>	Shared column utilities used by both CSV and Excel builders: <code>CHANNEL_KEYS</code> , <code>presentChannels()</code> , <code>buildHeaderLabel()</code> , <code>formatCellValue()</code> , <code>escapeCsvField()</code> ; <code>ExportFormat</code> type (<code>'csv'</code> <code>'xlsx'</code>)
<code>src/lib/export/csv.ts</code>	<code>buildCsv(activities)</code> — builds an RFC 4180 CSV string in memory
<code>src/lib/export/excel.ts</code>	<code>buildWorkbook(activities)</code> — builds a SheetJS workbook in memory and returns it as an <code>ArrayBuffer</code>
<code>src/lib/export/exportActivities.ts</code>	<code>exportActivities(activities, format)</code> — async entry point: lazy-imports <code>csv.ts</code> or <code>excel.ts</code> based on <code>format</code> (defaults to <code>'csv'</code>), then calls <code>triggerDownload</code>
<code>src/lib/export/download.ts</code>	<code>triggerDownload(data, filename, mime)</code> , <code>downloadPng(dataUrl, filename)</code> , <code>localDateString()</code>

Both `csv.ts` and `excel.ts` are lazy-imported in `exportActivities.ts` so that SheetJS and the CSV builder are only bundled into dynamically loaded chunks, keeping the main bundle lighter.

Shared column utilities (`columns.ts`)

`columns.ts` is the single source of truth for which channels appear in exports and how values are formatted. Both `csv.ts` and `excel.ts` import from it to prevent drift as new channels are added.

- `CHANNEL_KEYS` — ordered list of all exportable `ChannelKey` values
- `presentChannels(records)` — filters `CHANNEL_KEYS` to those with at least one non-null value across the provided records
- `buildHeaderLabel(key)` — returns the human-readable column header (e.g. `"Heart Rate (bpm)"`)
- `formatCellValue(key, value)` — formats a raw value for output: `pace` → `"M:SS"` string; timestamps → `Date` ; others → number or `null`
- `escapeCsvField(value)` — wraps a string in double-quotes and doubles any internal quotes per RFC 4180; returns the original string if no quoting is needed

CSV structure (`csv.ts`)

`buildCsv(activities)` produces an RFC 4180 string:

- **Single activity** — header row + one row per `ActivityRecord` ; no `activity` column
- **Multiple activities** — header row + all records from all activities in order; `activity` column (source filename) prepended as column 1
- Channels with all-null values across all records are omitted

- Pace formatted as "M:SS" ; timestamps as ISO 8601 strings
- Lines separated by CRLF; file ends with a trailing CRLF

Excel workbook structure (`excel.ts`)

`buildWorkbook` produces:

1. **Summary sheet** — one row per activity with filename, sport, start time (Excel datetime), distance (km), and elapsed time (H:MM:SS or M:SS).
2. **Per-activity sheets** — one row per `ActivityRecord`. Sheet names are taken from `activity.filename`, truncated to 31 characters (Excel limit), and de-duplicated with (N) suffixes if needed.

Column selection is dynamic: `presentChannels()` from `columns.ts` filters to channels with at least one non-null value in the activity.

Pace is formatted as an "M:SS" string by `formatCellValue()` rather than stored as a decimal, since Excel has no built-in pace format and decimal values are not human-readable.

Timestamps use JavaScript `Date` objects with `cellDates: true` in the SheetJS write options, with `yyyy-mm-dd hh:mm:ss` cell format applied to datetime columns.

PNG download

Each chart component calls `chart.getDataURL({ type: 'png', pixelRatio: 2, backgroundColor: bgColour })` on its ECharts instance, then passes the resulting base64 data URL to `downloadPng()`. `downloadPng` decodes the base64 string to a `Uint8Array`, wraps it in a `Blob`, and calls `triggerDownload` to initiate the browser download.

The background colour is read from the CSS custom property `--color-bg-primary` at download time so that the PNG matches the user's current theme.

`localDateString()`

`new Date().toISOString()` returns UTC, which rolls over to the next day before midnight for users in UTC+ timezones. `localDateString()` uses `toLocaleDateString('en-CA')` instead, which produces an ISO-format (`YYYY-MM-DD`) date in the user's local timezone.

3c. Chart Stats Row

PR #85 added a compact stats row below each `TimeSeriesChart` showing minimum, average, and maximum for every active series (added in PR #86). Stats update reactively when devices are toggled, and recalculate when the user zooms into a region.

Pace inversion (PR #88): Because pace is an inverted metric (lower min/km = faster), `computeSeriesStats()` swaps the min/max values for the `pace` channel so that `SeriesStats.max` holds the fastest (numerically smallest) pace and `SeriesStats.min` holds the slowest. The template renders pace as `max / avg / min` so numbers still read ascending left-to-right, consistent with the pace chart's inverted y-axis.

Module layout

File	Responsibility
<code>src/lib/components/charts/TimeSeriesChart.utils.ts</code>	<code>computeSeriesStats()</code> , <code>formatStatValue()</code> , <code>SeriesStats</code> interface
<code>src/lib/components/charts/TimeSeriesChart.svelte</code>	<code>zoomRange</code> state, <code>seriesStats</code> derived value, <code>.chart-stats</code> HTML/CSS

Key functions

`formatStatValue(value, channel)` — formats a raw number for display. Pace channels use `paceFormat()` (M:SS — re-exported from `$lib/utils/formatting` as the shared `formatPace`); integer channels (heartRate, power, cadence) round to zero decimal places; float channels (speed, temperature) show one decimal.

`computeSeriesStats(data, channel, label, colour, xRange?)` — slices `data` to the optional `xRange` window, passes y-values to `summarise()`, and returns a `SeriesStats` object (or `null` if no data). The `xRange` is provided by the zoom handler.

`SeriesStats` interface:

```
interface SeriesStats {
  label: string;
  colour: string;
  min: string; // formatted for display (pace: slowest; others: numeric minimum)
  avg: string; // formatted for display
  max: string; // formatted for display (pace: fastest; others: numeric maximum)
  minRaw: number;
  avgRaw: number;
  maxRaw: number;
  count: number;
  xMin: number;
  xMax: number;
  unit: string;
}
```

Zoom integration

`TimeSeriesChart.svelte` listens to ECharts' `dataZoom` event. When a zoom is active, the x-axis extent is read from ECharts' internal model via `chart.getModel().getComponent('xAxis', 0)` and stored in `zoomRange`. The `seriesStats` derived value re-runs whenever `zoomRange` changes, slicing each series' data to the visible window.

`zoomRange` is reset to `undefined` on every chart rebuild (new data, smoothing change, axis mode change), so stats always reflect the full dataset after a rebuild.

Data source vs Summary tab

The stats row and the Summary tab use the same `summarise()` function but different data sources:

Surface	Data source	Notes
Stats row	<code>buildData()</code>	Smoothed; distance-axis interpolated when in distance mode
Summary tab	<code>extractChannel(activity.records, ch)</code>	Raw unsmoothed records

When smoothing is 1 s and the axis is time, the two surfaces agree closely. In distance mode or with smoothing > 1 s, values may differ. Both surfaces carry a `title` tooltip explaining the difference.

3d. Activity Alignment — GPS Anchor and Indoor Detection

PR #91 replaced raw start-time-difference alignment with GPS-anchor-based alignment. PR #92 extended this to handle indoor activities (TrainerRoad, Zwift, treadmill) where GPS is absent or synthetic.

Indoor activity detection

`classifyIndoor(subSport, records)` in `src/lib/fit/parser.ts` determines `activity.isIndoor` at parse time using two signals:

1. **sub_sport field** (primary) — if the FIT session contains a recognised indoor `sub_sport` value, the activity is classified as indoor regardless of GPS presence. The full set is:

```
const INDOOR_SUB_SPORTS = new Set([
  'indoor_cycling', 'virtual_activity', 'spin', 'stationary_bike',
  'treadmill', 'indoor_rowing', 'indoor_running', 'indoor_walking',
  'virtual_ride', 'virtual_run',
]);
```

Key insight: Both Zwift (`virtual_activity`) and TrainerRoad (`indoor_cycling`) embed fake GPS coordinates in every record (Zwift uses virtual-world positions; TrainerRoad uses a Central Park, NYC placeholder). GPS presence cannot be used as the indoor detector — `sub_sport` is authoritative.

2. **GPS-absence fallback** — only fires when `sub_sport` is entirely absent (`undefined`) and all records lack a GPS position. Covers older devices that do not write `sub_sport` .

`activity.subSport` stores the raw `sub_sport` string from the FIT session (or `undefined`).
`activity.isIndoor: boolean` is always populated.

Anchor hierarchy

`findAnchor(activity)` in `src/lib/align/anchor.ts` branches on `activity.isIndoor`:

Outdoor path (unchanged from #91):

Priority	Source	AnchorSource
1	FIT timer-start event (within 30 s)	'timer'
2	First GPS record with speed > 0	'gpsMovement'
3	First GPS record (stationary)	'gpsFix'
4	File start	'fileStart'

Indoor path (added in #92 — GPS signals are skipped entirely):

Priority	Source	AnchorSource
1	FIT timer-start event (within 30 s)	'timer'
2	First <code>workout_step</code> timestamp (within 30 s)	'workoutStep'
3	First record with speed/power/cadence > 0	'indoorMovement'
4	File start	'fileStart'

The result is an `AlignmentAnchor` stored on `Activity` at parse time:

```
type AnchorSource = 'timer' | 'gpsMovement' | 'gpsFix' | 'fileStart'
                    | 'workoutStep' | 'indoorMovement';

interface AlignmentAnchor {
  recordIndex: number;
  distanceMetres: number;
  elapsedSeconds: number;
  timestamp: Date;
  source: AnchorSource;
}
```

`activity.anchor` is computed once in `parser.ts → normalise()` and available to all downstream code. The new `Activity` fields supporting indoor alignment are:

Field	Type	Description
subSport	string undefined	Raw FIT sub_sport value
isIndoor	boolean	Computed by classifyIndoor() at parse time
firstIndoorMovementIndex	number null	First record with speed/power/cadence > 0
firstWorkoutStepTime	Date null	Timestamp of first workout_step FIT message

Time-axis alignment

computeAnchoredOffsets(activities) in src/lib/align/timestamp.ts aligns files relative to the first loaded file's anchor timestamp rather than raw startTime values.

The TimeOffsetControl component shows a colour-coded anchor badge per file. Labels and tooltips are defined in TimeOffsetControl.utils.ts as ANCHOR_LABELS and ANCHOR_TITLES (both typed as Record<AnchorSource, string> for compile-time completeness):

Badge	Colour	Anchor source
Timer	Blue	'timer'
GPS move	Blue	'gpsMovement'
GPS fix	Amber	'gpsFix'
File start	Grey	'fileStart'
Workout	Green	'workoutStep'
Indoor	Green	'indoorMovement'

Distance-axis re-zeroing

interpolateToDistanceAxis(records, axis, distanceOffset) shifts the distance axis so that distanceOffset maps to 0 km. activity.anchor.distanceMetres is passed as distanceOffset in both Compare and Event pages.

Indoor/mixed session warnings — `createIndoorWarnings()`

src/lib/utils/indoorWarnings.svelte.ts exports a Svelte 5 runes composable that encapsulates the shared indoor warning state used by both Compare and Event pages:

```
const indoor = createIndoorWarnings(() => $activities);
// indoor.hasMixedIndoorOutdoor – any indoor + any outdoor file loaded (requires ≥2 files)
// indoor.allIndoor – all loaded files are indoor (single file or more)
// indoor.mixedWarningDismissed – dismissal state for amber banner
// indoor.indoorInfoDismissed – dismissal state for blue banner
```

The composable owns the dismissal `reset $effect` (fires on every `$activities` change). Each page adds two separate `$effect` blocks for axis auto-switching:

- **Mixed session:** `if (hasMixedIndoorOutdoor && $xAxisMode === 'distance')
xAxisMode.set('time')` — locks to Time because mixed distance axes are incompatible.
- **All-indoor session:** `if (allIndoor && untrack(() => $xAxisMode) === 'distance')
xAxisMode.set('time')` — defaults to Time on file load but uses `untrack()` so the user can manually switch back to Distance. `untrack()` prevents `$xAxisMode` from becoming a reactive dependency, limiting the effect to firing only when the file set changes.

CSS for the banners is in `src/routes/indoor-warning.css` (shared by both pages).

Proximity warning

`anchorsAreDistant(activities)` returns `true` when any two GPS-anchored activities have anchor positions more than `GPS_PROXIMITY_THRESHOLD_M` (50 m) apart. Both pages show a dismissible amber banner. Indoor activities whose anchors have no GPS position are excluded from the comparison, so a pure-TR file (no GPS) paired with a Zwift file never triggers a false proximity warning.

3e. Data Anomaly Detection

PR #147 added a data quality layer that detects sensor anomalies at parse time and surfaces them in the UI.

Detection engine — ``src/lib/analytics/anomalies.ts``

`detectAnomalies(records, options?)` is called once in `normalise()` and returns a sorted `Anomaly[]` stored on `Activity.anomalies`. Detection is $O(n)$ per channel.

Types and strategies:

```
type AnomalyType      = 'spike' | 'dropout' | 'gps-drift';
type DetectionStrategy = 'threshold-relative' | 'statistical';

interface Anomaly {
  channel:      ChannelKey;
  recordIndex: number;
  type:         AnomalyType;
  value:        number;      // raw sensor value at that record
  detectionStrategy: DetectionStrategy;
}
```

Spike detection — channels: `heartRate`, `power`, `cadence`

Condition	Threshold	Strategy
HR + <code>options.athleteProfile.maxHR</code> set	<code>maxHR + 10 bpm</code>	<code>threshold-relative</code>
Power + <code>options.powerSource === 'stryd' + cp</code> set	<code>cp × 1.5</code>	<code>threshold-relative</code>
Fallback (no profile)	<code>mean + 4σ</code>	<code>statistical</code>

A run above threshold must last ≤ 3 s to be flagged as a spike (longer sustained high values are real effort, not glitches).

Dropout detection — channels: `heartRate`, `power`, `cadence`

A channel value of `0` or `null` for more than 10 consecutive seconds mid-activity (excluding the first/last 30 s warmup/cooldown) is flagged as `type: 'dropout'`, `detectionStrategy: 'threshold-relative'`.

GPS drift detection — channel: `position`

Consecutive GPS positions are compared using the Haversine formula. If the implied speed between two points exceeds $3\times$ the recorded device speed, the record is flagged as `type: 'gps-drift'`, `detectionStrategy: 'threshold-relative'`. GPS drift anomalies use the `'position'` ChannelKey — a GPS-only channel with no numeric chart series. They surface only through the DeviceToggleBar GPS section and as red circle markers on the map.

Options pattern — `AnomalyDetectionOptions`:

```
interface AnomalyDetectionOptions {
  powerSource?: 'stryd' | 'native';
  athleteProfile?: { cp?: number; ftp?: number; maxHR?: number; };
}
```

Currently `detectAnomalies(records)` is called without options from `parser.ts` (statistical fallback for all channels). When issue #112 (athlete profile) and #117 (power source) ship, only the call site changes — the detection logic itself is unchanged.

Event grouping — `groupAnomalyEvents()`

`detectAnomalies` returns one `Anomaly` per record. A 40-second dropout produces 40 entries. `groupAnomalyEvents(anomalies)` collapses contiguous runs of the same channel + type (consecutive `recordIndex` values differing by ≤ 1) into a single entry (the first record of each run). This converts per-record counts to per-event counts for badge display.

```
// 40 dropout records → 1 dropout event
groupAnomalyEvents(activity.anomalies.filter(a => a.channel === 'heartRate'))
// → [Anomaly { recordIndex: startOfDropout, type: 'dropout', ... }]
```

Both Compare and Event pages call `groupAnomalyEvents()` before passing anomalies to `TimeSeriesChart` (one `markPoint` per event, not per record).

UI surfaces

DeviceToggleBar badges (`src/lib/components/ui/DeviceToggleBar.svelte`)

`buildAnomalyCounts(activities)` in `DeviceToggleBar.utils.ts` aggregates event-level counts across all activities and returns a `Map<ChannelKey, AnomalyCount>`. The badge `△ N` is rendered next to each channel label that has anomalies. Tooltip breaks down counts (e.g. "2 spikes, 1 dropout"). A standalone **GPS** section appears at the bottom when `anomalyCounts.get('position')` is non-zero — GPS drift anomalies are surfaced here rather than in any numeric channel group.

ChannelToggleBar badges (`src/lib/components/ui/ChannelToggleBar.svelte`)

The Event page passes `anomalyCounts` to `ChannelToggleBar`. Each channel pill shows an inline `△N` badge when that channel has anomaly events. No GPS section (Event page has no ActivityMap drift markers).

TimeSeriesChart markPoints (`src/lib/components/charts/TimeSeriesChart.svelte`)

`TimeSeriesChart` accepts `anomalies?: Anomaly[]` (pre-filtered to the chart's channel and pre-grouped via `groupAnomalyEvents`). A red diamond `markPoint` is rendered at each anomaly's x-axis position on the **first visible series**. Positions use `anomalyXValue()` from `TimeSeriesChart.utils.ts` — which returns `elapsedSeconds` in time mode and kilometres in distance mode. Clicking a `markPoint` zooms the chart to ± 15 s (time mode) or ± 50 m (distance mode) around the anomaly.

ActivityMap drift circles (`src/lib/components/map/ActivityMap.svelte`)

`ActivityMap` reads `activity.anomalies` directly (no separate prop). GPS-drift anomalies are filtered by `a.type === 'gps-drift'`, and each anomaly's corresponding `ActivityRecord.position` is rendered as a red `L.circleMarker` (radius 5, `#ef4444`). Non-GPS anomalies are ignored on the map.

`position` ChannelKey

`'position'` was added to the `ChannelKey` union to give GPS drift anomalies a first-class channel assignment. It does NOT appear in `ALL_RECORD_CHANNELS` (the ordered list used by `channelsPresentInRecords`), so it is never returned by `deriveAvailableChannels()` and never rendered as a chart series. Its sole purpose is to carry GPS drift anomaly data through `buildAnomalyCounts` to the DeviceToggleBar GPS section.

4. Responsive Layout

The app is fully responsive across three viewport tiers. Breakpoints are defined in two places that must stay in sync:

- **CSS:** `--bp-phone: 480px` and `--bp-tablet: 768px` in `src/routes/layout.css`
- **JS:** `PHONE_MAX = 480` and `TABLET_MAX = 768` in `src/lib/stores/viewport.ts`

The `viewport` readable store emits `'phone' | 'tablet' | 'desktop'` via `window.matchMedia` listeners. It is SSR-safe (defaults to `'desktop'`).

Key responsive components:

Component	Behaviour change at ≤768px
<code>Sidebar.svelte</code>	Switches to <code>position: fixed</code> slide-out drawer
<code>+layout.svelte</code>	Renders hamburger button (fixed, top-left, 44×44px) and backdrop
<code>CollapsiblePanel.svelte</code>	Wraps toolbar controls; <code>display: contents</code> on desktop, collapsible on mobile
<code>+layout.svelte</code> (phone only)	Renders bottom navigation bar at ≤480px

5. FIT Parsing

FIT files are parsed by `fit-file-parser` running in a **Web Worker** to avoid blocking the UI thread. The worker is created in `src/lib/fit/parser.ts` .

Key parsing details:

- `STRING_DEVICE_TYPE` mapping converts `fit-file-parser` 's `snake_case` string outputs (e.g. `'heart_rate'`) back to numeric ANT+ constants.
- Device deduplication: duplicate `device_index` entries are discarded (only first occurrence retained).
- Two-pass channel allocation: external sensors (known ANT+ types) claim channels first; the watch claims remaining channels; any still-unclaimed channels are merged into the first device as a safety net.
- Enhanced fields: `enhanced_speed` and `enhanced_altitude` are preferred over `speed` and `altitude` .
- Stryd developer fields: `Power` (capital P), `stance_time` , `step_length` are mapped to standard channel keys.

Sport-specific post-processing is applied after record normalisation:

Sport	Function	Reason
<code>running</code>	<code>applyRunningCadenceDoubling(records)</code>	FIT cadence is single-leg (one foot per minute). Doubles to get conventional spm (steps per minute).
<code>cycling</code>	<code>removeCyclingPace(records)</code>	Pace (min/km) is not a meaningful cycling metric. Clears all <code>pace</code> values so the channel is excluded from device streams and charts entirely.

Both functions are exported from `parser.ts` for unit testing.

Record validation is applied in `normalise()` after record normalisation and before sport-specific post-processing (PR #142):

Function	Behaviour	Toast / log
<code>filterNegativeElapsed(records)</code>	Returns a new array with all records where <code>elapsedSeconds < 0</code> removed. Broken device clocks can emit these.	<code>console.warn</code> + 'warning' toast if any are removed.
<code>ensureSortedByElapsed(records)</code>	Sorts <code>records</code> in-place by <code>elapsedSeconds</code> if out of order; returns <code>true</code> when sorting was required.	Caller (<code>normalise()</code>) emits <code>console.warn</code> + 'warning' toast on <code>true</code> .

Both functions are exported from `parser.ts` for unit testing. Well-formed files pass through both with no output.

Lap building is handled by `buildLaps(fitLaps, records)` , which is also exported for direct unit testing. It derives `startDistance` and `endDistance` from the running cursor position in the records array rather than from the FIT lap's `start_distance` field. This makes it resilient to Garmin devices that write `start_distance = 0` for every lap (per-lap relative) rather than cumulative from the session start. Laps with `total_distance = 0` are returned as zero-length entries (`startIndex === endIndex` , `startDistance === endDistance`) without advancing the cursor.

The cursor advancement uses `records[cursor].distance <= targetDist + DISTANCE_EPSILON_M` (where `DISTANCE_EPSILON_M = 0.5`) to absorb GPS floating-point drift at lap boundaries. The constant is exported for use in tests and any future callers that need the same tolerance.

Anomaly detection is called at the end of `normalise()` after sport-specific post-processing and before `buildLaps()` . See section 3e for details.

Available channel pre-computation — `channelsPresentInRecords(records)` scans each `ActivityRecord` once across all 14 channel keys and returns a `Set<ChannelKey>` of those with at least one non-null value. It is called once inside `normalise()` and the result is stored on `Activity` as `availableChannels: Set<ChannelKey>` (PR #108).

`deriveAvailableChannels(activities)` in `src/lib/utils/channels.ts` unions those pre-computed sets rather than re-scanning records on every render call. It returns the channel list in the canonical ordering defined by `ALL_RECORD_CHANNELS` (exported from `parser.ts`), which also replaces the formerly duplicated `ALL_CHANNELS` constant that used to live only in `channels.ts` .

6. Testing

Unit tests use **Vitest**. Test files live alongside the modules they test (e.g. `TimeSeriesChart.test.ts` next to `TimeSeriesChart.svelte`).

```
npm test          # run all tests
npm run test:watch # watch mode
```

Test environments

Most tests run in the default `node` environment (pure utility functions with no DOM). Component tests that mount Svelte components require a DOM and use `jsdom`.

To mark a test file as a component test, add this directive as the **first line** of the file:

```
// @vitest-environment jsdom
```

Why the `browser` resolve condition is needed for component tests:

Svelte 5 exports both a browser build (`index.js`) and a server build (`index-server.js`). In `node` environment, Vite resolves to the server build, which throws `"mount(...) is not available on the server"`. The `vite.config.ts` sets `resolve.conditions: ['browser']` **only when** `mode === 'test'` so that `jsdom` component tests use the browser build. The production build is unaffected (Vite's built-in defaults include `browser` for client bundles).

```
// vite.config.ts – mode-conditional resolve
export default defineConfig(({ mode }) => ({
  plugins: [tailwindcss(), sveltekit(), visualizer({ filename: 'stats.html', open: false }),
  build: { chunkSizeWarningLimit: 1200 }, // ECharts async chunk is ~1.1 MB – suppress
  ...(mode === 'test' ? { resolve: { conditions: ['browser'] } } : {}),
  // ...
}));
```

Mocking Leaflet in component tests:

`ActivityMap.svelte` dynamically imports Leaflet in `onMount`. In `jsdom`, Leaflet's browser DOM APIs are unavailable. Mock Leaflet at the module level:

```
vi.mock('leaflet', () => { /* minimal stubs for map, tileLayer, control, etc. */ });
vi.mock('leaflet/dist/leaflet.css', () => ({}));
```

Also stub `ResizeObserver` (not implemented in `jsdom`):

```
vi.stubGlobal('ResizeObserver', class { observe = vi.fn(); disconnect = vi.fn(); });
```

There are no integration or end-to-end tests at present. The API routes for sync are exercised manually — see section 3.11 for local setup.

Shared test helper — `makeBaseActivity``

`src/lib/test-utils.ts` exports `makeBaseActivity(overrides?)`, a single factory for building `Activity` fixture objects in tests (PR #108):

```
import { makeBaseActivity } from '$lib/test-utils';

const activity = makeBaseActivity({ id: 'run-1', records: [/* ... */] });
```

The factory fills every required `Activity` field with safe defaults and computes `availableChannels` automatically via `channelsPresentInRecords(records)`. It also sets `anomalies: []` as the default empty anomaly list, matching `normalise()`'s initial state before `detectAnomalies()` is called. Pass `overrides.availableChannels` to supply a manual set when the test is specifically about channel visibility rather than parser behaviour.

All test files should import this helper rather than duplicating the full `Activity` literal — it prevents future interface changes from requiring edits across every test file.